I)

$$A\underline{x} = \lambda \underline{x}$$

<u>Self-adjoint</u>

$$A = \underset{\sim}{U} \Lambda \underset{\sim}{U}^H$$

U: unitary

$$\Lambda = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_m & & \\ & & \ddots & \\ & & & \lambda_m \end{bmatrix}$$

$$= \sum_{i=1}^{m} \lambda_i \, \underline{y}_i \underline{y}_i^H$$

$$\underbrace{[\,|\,][-]}$$

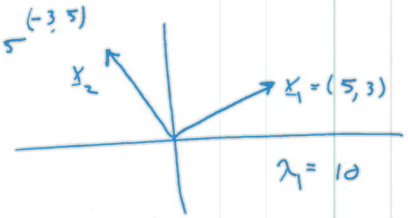
x_i : eigenvector

λ_i : eigenvalue

Suppose we want to build a matrix with eigenvalues λ_i and eigenvectors x_i.

want R:

$$\underline{x}_2 = \frac{1}{\sqrt{34}}\begin{bmatrix} -3 \\ 5 \end{bmatrix} \qquad \lambda_2 = 5 \quad (-3, 5)$$



$$\underline{x}_1 = \frac{1}{\sqrt{34}}\begin{bmatrix} 5 \\ 3 \end{bmatrix} \qquad \lambda_1 = 10$$

$$R = 10\left(\frac{1}{34}\right)\begin{bmatrix} 5 \\ 3 \end{bmatrix}\begin{bmatrix} 5 & 3 \end{bmatrix} + 5\left(\frac{1}{34}\right)\begin{bmatrix} -3 \\ 5 \end{bmatrix}\begin{bmatrix} -3 & 5 \end{bmatrix}$$

II)

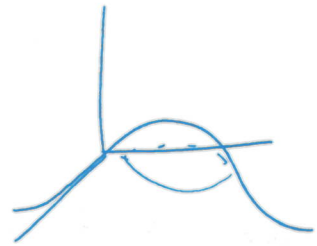
$$X \sim N(\underline{\mu}, R)$$

~ PDF

The ellipsis is oriented along the eigenvectors of the covariance matrix R.

$$f(\underline{x}) = (k) \exp\left[-\frac{1}{2}(\underline{x}-\underline{\mu})^T R^{-1} (\underline{x}-\underline{\mu})\right]$$

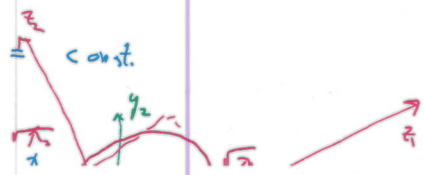
want plot $\underline{x}$ s.t. $f(\underline{x}) =$ constant.
(contour plot)



$f(\underline{x}) =$ cons.

$$\log f(x) \longrightarrow (\underline{x}-\underline{\mu})^T R^{-1} (\underline{x}-\underline{\mu}) = \text{const.}$$

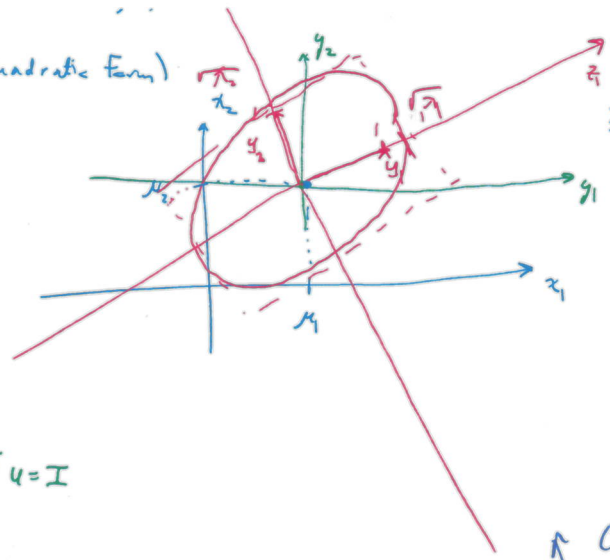
(Quadratic Form)



$[x.]$

This causes a shift in the axis (now we have y_1 y_2)

(Quadratic form)



$$y = x - \mu$$

Change of variables

$$y^T R^{-1} y$$

Do an eigen decomposition

$$R = U \Lambda U^T$$

$$U = [y_1 \quad y_2] \qquad U^T U = I$$

$$\Lambda = \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix}$$

coordinate directions of red axis

Change of variables (again)

$$z = U^T y \qquad y = U z$$

$$R^{-1} = U \Lambda^{-1} U^T$$

$$y^T R^{-1} y = z^T \cancel{U^T} \cancel{U} \Lambda^{-1} \cancel{U^T} \cancel{U} z$$

$$= z^T \Lambda^{-1} z$$

$$= [z_1 \quad z_2] \begin{bmatrix} 1/\lambda_1 & \\ & 1/\lambda_2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

$$= \frac{z_1^2}{\lambda_1} + \frac{z_2^2}{\lambda_2} = \text{const.}$$

Suppose $z = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

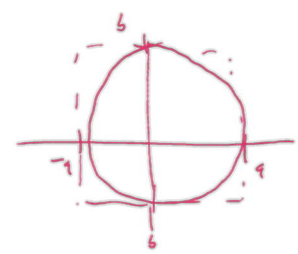
$y = Uz = y_1$

$z = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$y = Uz = y_2$

set z to the unit directions to get eigenvector directions

$$\frac{x^2}{4^2} + \frac{y^2}{3^2} = 1$$



A)

Rayleigh Quotient

$$R_A(x) = \frac{x^H A x}{x^H x}$$

only the direction matters

A: self adjoint

What we want to do — $\max_{x \neq 0} R_A(x)$

We can say this because the norm of x does not matter, just the direction — $\max_{\|x\|_2 = 1} R_A(x)$

$$\max \; x^H A x \quad \text{s.t.} \quad x^H x = 1$$

1)

Given (1) how can I use that to get (2)?

$$A\underline{x} = \lambda \underline{x} \qquad (1)$$

$$(A + \sigma^2 I)\underline{y} = \mu \underline{y} \qquad (2)$$

$$A\underline{y} + \sigma^2 \underline{y} = \mu \underline{y}$$

$$\boxed{A\underline{y} = (\mu - \sigma^2)\underline{y}}$$

Adding on $\sigma^2 I$ does not change the eigenvectors so $y = x$.

$$\underline{y} = \underline{x}$$

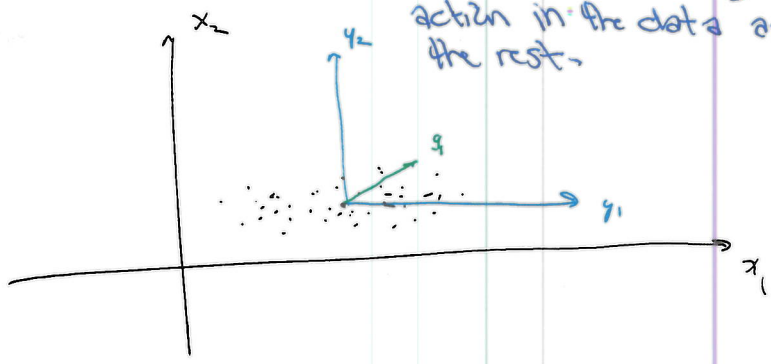
$$\mu - \sigma^2 = \lambda$$

$$\boxed{\mu = \lambda + \sigma^2}$$

(eigenvectors of $A + \sigma^2 I$ are the as for A)

B)

Principal Components – The purpose is to get high dimensional data and reduce it to smaller dimension data in such a way that captures the action in the data and throws away the rest.



Data: $\underline{x}_1, \underline{x}_2, \ldots, \underline{x}_N$, $\underline{x}_i \in \mathbb{R}^d$

Mean: $\frac{1}{N}\sum_{i=1}^{N} \underline{x}_i = \bar{\underline{x}}$

Offset the axis to the mean of the data

$$\underline{y}_i = \underline{x}_i - \bar{\underline{x}}$$

Direction vectors: $\underline{q}_1$: $\|\underline{q}_1\| = 1$ $\quad \underline{q}_1^T \underline{q}_1 = 1$

component of $\underline{y}_i$ in the $\underline{q}_1$ direction is

$$\langle \underline{y}_i, \underline{q}_1 \rangle = \underline{q}_1^T \underline{y}_i$$

From $J = \frac{1}{N}\sum^{N} (a^T u_i)^2$

Form $J = \frac{1}{N} \sum_{i=1}^{N} (q_1^T y_i)^2$

Goal: choose q_1 to make J as large as possible.

Estimate of the covariance matrix — let $S = \underbrace{\frac{1}{N} \sum_{i=1}^{N} y_i y_i^T}$ (scatter matrix)

Expand J — $J = \frac{1}{N} \sum_{i=1}^{N} q_1^T y_i y_i^T q_1$

$$= q_1^T \frac{1}{N} \left[\sum_{i=1}^{N} y_i y_i^T \right] q_1$$

$$= q_1^T S q_1$$

Create the constrained equation problem

$$\max \quad q_1^T S q_1$$
$$\text{s.t.} \quad q_1^T q_1 = 1$$

Rayleigh Quotient Problem!